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Writing parametric flood triggers from a global flood model: a gauge-by-gauge test in the Guadalquivir basin, Spain

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ABSTRACT

A parametric flood policy pays when a river index crosses an agreed trigger. Where a river has no gauge, the index has to come from a global flood model, and the policy is only as good as the model's agreement with the floods that actually happen. We test how a trigger set from the GloFAS reanalysis performs against 14 river gauges in the Guadalquivir basin, Spain (583–16,166 km², water years 1985–2022, 286 gauge-years). Flood-frequency curves were fitted to annual maxima from both sources by L-moments (GEV and log-Pearson III), and 1-in-T triggers were scored year by year against the floods the gauges measured. The model's 10-year flood is 0.22 of the gauged one at the median gauge, so a trigger written in m³/s from the model caught only 6 of 60 1-in-5 flood years (Peirce skill 0.06, no better than chance). Written instead as the model's own 1-in-5 level, the same trigger caught 30 of 60 (hit rate 0.50, false-alarm rate 0.12, Peirce skill 0.38); 24 flood years were caught only by the return-period trigger and 0 only by the fixed-flow one (McNemar $p = 1.2 \times 10^{-7}$). Basis risk remained high (27 false payouts, 47% of all payouts). Across gauges, the rank correlation of annual floods between gauge and model predicted the share of flood years caught (Spearman $\rho = 0.80$, $p = 0.0006$), which gives a simple screen for basis risk wherever a short local record exists. Triggers from a global model should be written as the model's own return period and calibrated locally before they enter a contract. Data, code and a notebook that reproduces every number are published with this paper.

Keywords: parametric insurance; flood trigger; basis risk; GloFAS; flood frequency; L-moments; verification; Guadalquivir

1. Introduction

Parametric (index) insurance pays a fixed sum when a measured index passes an agreed level, instead of after a loss assessment. Payouts are fast and cheap to administer, which is why the approach has spread from crop and weather cover to flood risk, risk pools and forecast-based humanitarian finance [4]. Its weakness is basis risk: the index can pass the trigger when the insured suffers no loss, or stay below it when they do. Basis risk is the main reason that demand for index cover stays lower than its price alone would suggest [3].

For river floods the natural index is discharge at a gauge. Most places at risk have no gauge, and where one exists its record is often too short, interrupted or not public. Global hydrological reanalyses fill the gap. The Global Flood Awareness System (GloFAS) runs the LISFLOOD model over the whole land surface at about 5 km resolution, driven by atmospheric reanalysis, and publishes daily river discharge from 1979 onwards [1, 7]. A trigger for any river reach can therefore be read from GloFAS in minutes. The question a buyer and an insurer must answer first is whether that trigger pays in the years the river actually floods.

Global models are calibrated against a subset of the world's gauges [8] and are known to struggle where rivers are regulated by reservoirs and abstractions, and in semi-arid climates where runoff is a small and variable share of rainfall. Both conditions describe the Guadalquivir in southern Spain. This paper asks three practical questions there: (1) how close the model's flood magnitudes are to the gauged ones; (2) whether a trigger written as a fixed flow, or as the model's own return period, pays in the right years; and (3) whether the basis risk at a site can be anticipated before a policy is written. We answer them with an open, reproducible test on 14 gauges.

2. Data

Gauges. Daily mean river flow and station metadata come from the *Anuario de Aforos 2021-2022* for the Guadalquivir basin, published by CEDEX for Spain's Ministry for the Ecological Transition [2]. We kept every gauge with a catchment of at least 500 km² and at least 25 years of daily flow in water years 1985–2022 (October to September). This gave 14 gauges with catchments of 583 to 16,166 km² (Table 1; Figure 1). The raw CEDEX files are not redistributed; they are fetched from CEDEX and checked against published SHA-256 checksums.

Global model. Daily river discharge (24-hour mean) from the GloFAS reanalysis, LISFLOOD, consolidated product, version 4.0, at 0.05° resolution, retrieved through the Open-Meteo flood API, which serves the Copernicus Emergency Management Service data under CC BY 4.0. For each gauge, all cells within ±0.1° (about 10 km) were retrieved, and the cell whose mean flow over the common period was closest to the gauge's (in log ratio) was used. This choice uses the gauge and is therefore the model's best case; at an ungauged site the cell has to be found from the upstream area alone.

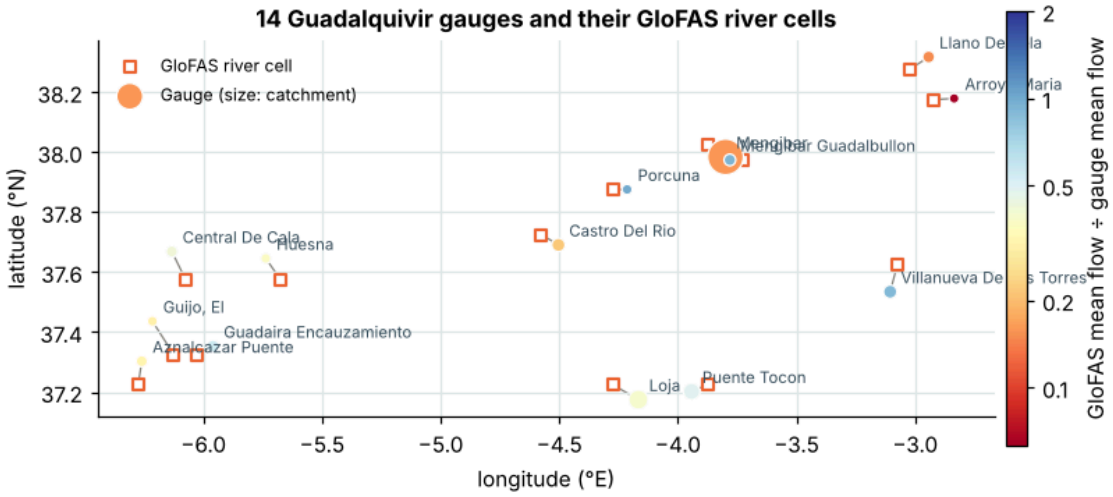


Figure 1. The 14 gauges (dots, sized by catchment area, coloured by the ratio of GloFAS mean flow to gauged mean flow) and their matched GloFAS river cells (squares). An interactive version with basemaps, rivers and sub-catchments is at https://kai-water.charkaks0.workers.dev/case-guadalquivir.

3. Methods

3.1 Annual maxima and flood frequency

For each source, the largest daily flow of each water year was taken, keeping years with at least 80 % of days; the comparison uses the years both sources cover (17 to 24 per gauge, 286 gauge-years in total). The generalised extreme value (GEV) distribution and the log-Pearson type III (LP3) distribution were fitted by L-moments [9, 10], the latter to $\log_{10}$ flows as in flood-frequency practice [5]. The better fit was chosen by the probability-plot correlation with Gringorten plotting positions [6]; the GEV was chosen at 11 of 14 gauges and at 10 of 14 GloFAS cells (LP3 is not fitted where a year's maximum is zero). Return levels for 1-in-2 to 1-in-50 years carry a 90 % percentile bootstrap band (300 resamples of the annual maxima, refitted).

3.2 Triggers

A year is a *flood year* at a gauge when its annual maximum reached the gauge's own fitted 1-in-T level. Two triggers were set from the model for each T = 2, 5, 10 and 25 years. The *fixed-flow* trigger pays when the model's annual maximum reaches the gauge's 1-in-T level in m³/s: the trigger a buyer would write by reading a design flow off a gauge-based study and applying it to the model. The *return-period* trigger pays when the model's annual maximum reaches the model's own 1-in-T level, i.e. the trigger is defined by frequency rather than by magnitude. Payouts are counted per gauge-year: a *hit* (flood year, paid), a *miss* (flood year, not paid) or a *false payout* (paid, no flood year).

3.3 Verification

From the pooled 2 × 2 table of all gauge-years we report the hit rate H, the false-alarm rate F (false payouts as a share of non-flood years), the share of payouts that were false, and the Peirce skill score H − F [13], which is 0 for a trigger with no skill and 1 for a perfect one, whatever the base rate. Against chance, we test whether a trigger that paid out the same number of times at random would have caught as many flood years (one-sided hypergeometric test). The two triggers are compared on the same flood years with McNemar's test [12] on the flood years caught by one but not the other. Across gauges, Spearman's rank correlation relates basis risk to the agreement between the two annual-maximum series.

3.4 Reproducibility

The analysis was run by the KAI engine and reproduced independently by an open notebook built on textbook formulas (NumPy, SciPy, pandas). The notebook re-derives every trigger count exactly and the median magnitude ratio to within 0.02 of the engine's results, and stops with an error if they drift. A rebuild script regenerates the derived data files from the CEDEX and GloFAS sources; it reproduced all 286 annual maxima and all 14 cell choices exactly.

4. Results

4.1 The model sees these floods at about a fifth of their size

GloFAS mean flow was within a factor of 2 of the gauge at only 4 of 14 gauges (median ratio 0.38), and the model's 10-year flood was 0.22 of the gauged one at the median gauge, within a factor of 2 at 4 gauges (Figure 2; Table 1). The low bias held at every return period (median ratios 0.21, 0.19, 0.22, 0.29 and 0.36 for 1-in-2 to 1-in-50). It did not depend on catchment size (Spearman ρ = 0.22, p = 0.44) but did follow the mean-flow bias (ρ = 0.54, p = 0.049): where the model's water balance is low, so are its floods. A search up to 25 km around the largest gauge (Mengibar, 16,166 km²) found no river cell with a larger flow, so the bias is in the model, not in the choice of cell.

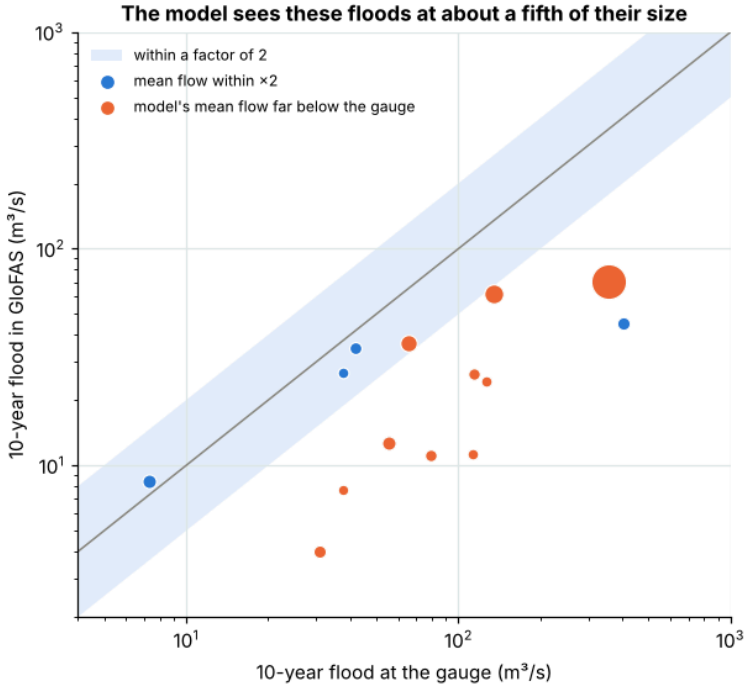


Figure 2. The 10-year flood at each gauge against the 10-year flood in GloFAS (log scales; dot size: catchment area). Blue: GloFAS mean flow within a factor of 2 of the gauge; orange: far below. Shading: within a factor of 2 of the 1 : 1 line.

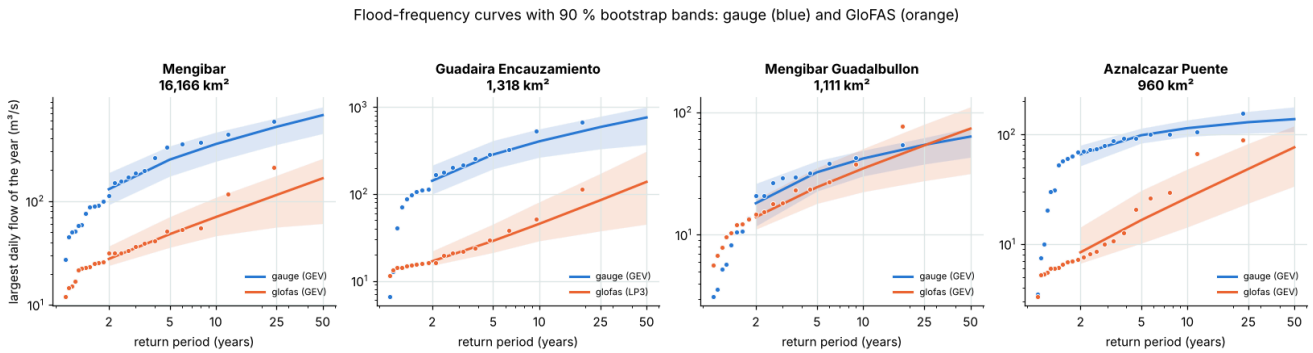


Figure 3. Flood-frequency curves at four gauges: annual maxima at their plotting positions, the preferred fit and its 90 % bootstrap band, for the gauge (blue) and GloFAS (orange). Where the curves run roughly parallel (Mengibar), a trigger written as a return period can still work; where the model is flatter or noisier, it cannot.

4.2 Write the trigger as a return period, not in m³/s

Because the model's floods are too small, a trigger written in m³/s from a gauge-based design flow almost never fired: at 1-in-5 it caught 6 of 60 flood years, with a Peirce skill of 0.06, no better than a trigger paying at random ($p = 0.093$; Table 2). Written as the model's own 1-in-5 level, the same trigger caught 30 flood years with a false-alarm rate of 0.12 (Peirce skill 0.38; $p = 1.4 \times 10^{-9}$ against chance). On the same flood years, 24 were caught by the return-period trigger but not the fixed-flow one, and 0 the other way round (McNemar $p = 1.2 \times 10^{-7}$). The ranking held at every trigger level (Figure 4): skill peaked at 1-in-5 (0.38) and fell to 0.22 at 1-in-25, where only 11 flood years exist in the pooled record. The return-period trigger is not free of basis risk: at 1-in-5 it also paid 27 times without a flood, so 47% of its payouts would have been false.

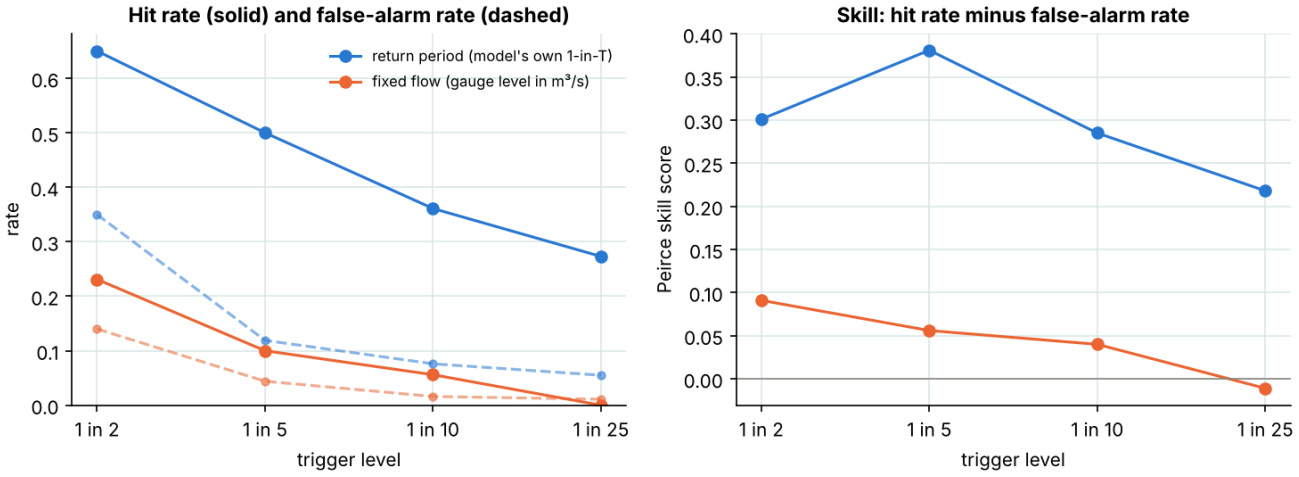


Figure 4. Pooled verification of the two triggers at four trigger levels: hit rate and false-alarm rate (left) and the Peirce skill score (right). The return-period trigger keeps a positive skill at every level; the fixed-flow trigger has almost none.

Table 2. Verification of 1-in-T triggers set from GloFAS, pooled over 14 gauges and 286 gauge-years.

Trigger	Written as	Caught	Missed	False payouts	Hit rate	False-alarm rate	Peirce skill	Caught by chance	p (vs chance)
1 in 2	Return period	93	50	50	0.65	0.350	0.30	71.5	2.8×10^{-7}
1 in 2	Fixed m³/s	33	110	20	0.23	0.140	0.09	26.5	0.034
1 in 5	Return period	30	30	27	0.50	0.119	0.38	12.0	1.4×10^{-9}
1 in 5	Fixed m³/s	6	54	10	0.10	0.044	0.06	3.4	0.093
1 in 10	Return period	13	23	19	0.36	0.076	0.28	4.0	1.6×10^{-5}
1 in 10	Fixed m³/s	2	34	4	0.06	0.016	0.04	0.8	0.17
1 in 25	Return period	3	8	15	0.27	0.055	0.22	0.7	0.025
1 in 25	Fixed m³/s	0	11	3	0.00	0.011	-0.01	0.1	1

Caught by chance: expected hits of a trigger that paid the same number of times at random; p: one-sided hypergeometric probability of at least as many hits by chance.

4.3 Basis risk can be anticipated from a short record

Skill varied strongly between gauges (Table 1). The share of 1-in-5 flood years caught ranged from 0.20 to 1.00, and it followed the rank correlation of the gauge's annual maxima with the model's (Spearman $\rho = 0.80$, $p = 0.0006$; Figure 5). The median rank correlation was 0.42. Gauges where the model ranked the big years well ($\rho \geq 0.6$: Mengibar, Castro del Río, Central de Cala, Aznalcázar Puente) caught half or more of their flood years, up to all of them; gauges where it did not (ρ near zero or negative: Villanueva de las Torres, Llano de Vela, Arroyo María) caught a fifth to two-fifths.

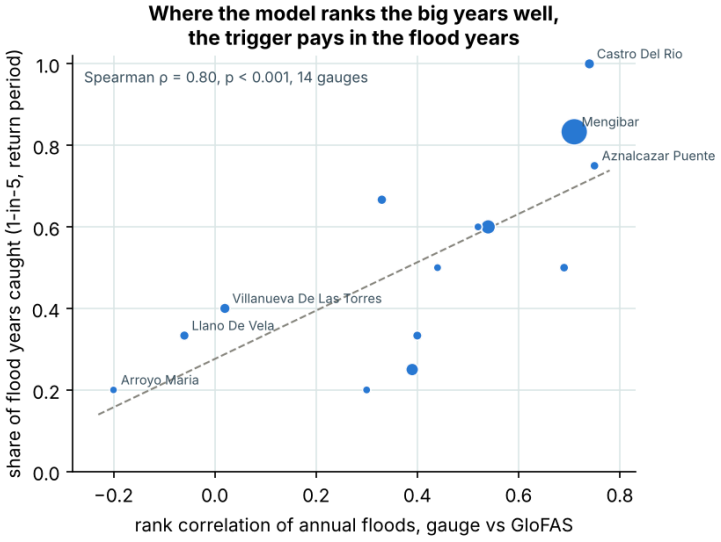


Figure 5. Share of 1-in-5 flood years caught by the return-period trigger at each gauge against the rank correlation of annual maxima, gauge against GloFAS (dot size: catchment area; dashed: least-squares line).

Table 1. Gauges, 10-year floods (Q10) with 90 % bootstrap bands, and the 1-in-5 return-period trigger at each gauge.

Gauge	Catchment km²	Years	Mean-flow ratio	Q10 gauge, m³/s (90 % band)	Q10 GloFAS, m³/s (90 % band)	Q10 ratio	Rank corr.	Flood years	Caught	False payouts
Mengibar	16,166	23	0.16	355 (242–460)	70 (46–109)	0.20	0.71	6	5	0
Loja	4,210	23	0.39	135 (88–181)	62 (48–78)	0.46	0.54	5	3	2
Puente Tocon	2,944	18	0.48	66 (42–86)	37 (29–42)	0.56	0.39	4	1	3
Villanueva De Las Torres	1,685	21	0.90	7 (5–10)	9 (6–10)	1.16	0.02	5	2	3
Castro Del Rio	1,665	17	0.22	56 (31–84)	13 (6–25)	0.23	0.74	2	2	1
Guadaira Encauzamiento	1,318	18	0.56	404 (264–560)	45 (29–81)	0.11	0.33	3	2	2
Llano De Vela	1,240	20	0.15	31 (24–37)	4 (3–5)	0.13	-0.06	3	1	2
Mengibar Guadalbullon	1,111	17	0.93	42 (30–49)	35 (22–50)	0.83	0.40	3	1	2
Central De Cala	1,010	19	0.42	79 (26–121)	11 (7–16)	0.14	0.69	6	3	1
Aznalcazar Puente	960	22	0.32	114 (96–135)	26 (14–48)	0.23	0.75	4	3	2
Huesna	678	21	0.37	126 (59–216)	24 (10–46)	0.19	0.52	5	3	2
Guijo, El	672	23	0.31	113 (74–146)	11 (9–13)	0.10	0.44	4	2	2
Porcuna	670	20	0.99	38 (20–54)	27 (14–46)	0.71	0.30	5	1	2
Arroyo Maria	583	24	0.05	37 (30–47)	8 (5–12)	0.21	-0.20	5	1	3

Mean-flow ratio: GloFAS mean flow ÷ gauged mean flow over the common days. Q10 ratio: GloFAS ÷ gauge. Rank corr.: Spearman correlation of annual maxima. Years: water years both sources cover.

5. Discussion

Frequency, not magnitude. A global model can be biased in magnitude and still carry useful information on which years are large. Writing the trigger as the model's own return period uses that information and discards the bias; writing it in m³/s keeps the bias and discards the information. In this basin the difference was between a trigger with clear skill and one with none. The same holds for a trigger written as a percentile of the model's flows, or as a share of a named past flood in the model, which are also defined in the model's own terms.

Basis risk remains, and can be screened. Even the better trigger paid without a flood in 47% of payouts at 1-in-5. For a buyer that is a large risk, and in a regulated basin it is expected: reservoirs cut and delay flood peaks in some years and not in others, which a model without the reservoirs' operation cannot reproduce. The strong link between rank correlation and the share of floods caught offers a practical screen: a few years of local evidence (a short gauge record, flood marks, satellite flood extents or insurance claims) give a rank correlation that indicates, before the policy is written, whether the global index can carry the trigger or needs local calibration.

Limits. One basin, strongly regulated and semi-arid, so the size of the bias will differ elsewhere. GloFAS 4.0; version 5.0 of the reanalysis is the next test. Daily means, so instantaneous peaks are higher than the values used. 17 to 24 years per gauge, so levels beyond 1-in-25 are extrapolations and the 1-in-25 scores rest on 11 flood years. The cell was matched with the gauge's mean flow, the model's best case. Flood years are defined from the gauges' daily flows, not from losses; a contract would also have to consider where and how losses occur.

6. Conclusions

On 14 gauges in the Guadalquivir basin, the GloFAS reanalysis saw floods at about a fifth of their measured size. A parametric trigger written in m³/s from the model was no better than chance; the same trigger written as the model's own 1-in-5 level caught half the flood years with a false-alarm rate of 0.12, but a large share of its payouts were still false. Three recommendations follow for anyone writing triggers from a global model: write them in the model's own terms (return period, percentile or share of a reference flood), never as a flow read from another source; screen basis risk with the rank correlation against any local record before the policy is written; and calibrate locally before the trigger enters a contract.

Data and code availability

The data package published with this paper on Zenodo contains the derived data (annual maxima at the gauges and at their GloFAS cells; gauge list), the result files, the figures, the notebook that reproduces every number, the shared functions, the script that rebuilds the derived data from the sources, and the scripts that compute the statistics quoted here and build this paper from them. The same material is maintained at https://github.com/Kaitale/kai-case-studies/tree/main/spain-guadalquivir-flood-triggers (code MIT; text, figures and derived data CC BY 4.0). An interactive version with maps is at https://kai-water.charkaks0.workers.dev/case-guadalquivir.

Acknowledgements and competing interests

Gauge data: *Fuente: CEDEX, Anuario de Aforos 2021-2022 (MITECO)*. Global model: contains modified Copernicus Emergency Management Service information 2026 (GloFAS 4.0), served by Open-Meteo (CC BY 4.0). Sub-catchments in the interactive maps: HydroBASINS [11]. The author founded KAI Water & Energy, which offers parametric trigger studies (KAI-Hydro). Analysis code and text were prepared with the help of an AI assistant (Claude, Anthropic) and checked by the author, who takes responsibility for the content.

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